Problem 3: Merge Two Sorted Lists (easy)

Problem Statement

Given the head of two sorted linked lists, `l1` and `l2`.

Return a new sorted list created by merging together the nodes of the first two lists.

Examples

1. **Example 1**:
   - **Input**:

     ```
     [1, 3, 5]
     [2, 4, 6]
     ```

   - **Expected Output**:

     ```
     [1, 2, 3, 4, 5, 6]
     ```

   - **Justification**: Merging the two sorted linked lists, node by node, results in a single sorted linked list containing all elements from both input lists.
2. **Example 2**:
   - **Input**:

     ```
     [2, 4, 6]
     [1, 3, 5]
     ```

   - **Expected Output**:

     ```
     [1, 2, 3, 4, 5, 6]
     ```

   - **Justification**: Both lists are in ascending order; merging them node by node in ascending order gives us the sorted linked list with all elements.
3. **Example 3**:
   - **Input**:

     ```
     [1, 2, 3]
     [4, 5, 6]
     ```

   - **Expected Output**:

     ```
     [1, 2, 3, 4, 5, 6]
     ```

   - **Justification**: As the first list contains all smaller elements, combining them results in a new list with elements from the first list followed by the second one.

Try it yourself

Try solving this question here:

Java

```java
// class ListNode {
//     int val;
//     ListNode next;
//     ListNode() {}
//     ListNode(int val) { this.val = val; }
//     ListNode(int val, ListNode next) { this.val = val; this.next = next; }
// }

public class Solution {
    // Function to merge two sorted linked lists
    public ListNode mergeTwoLists(ListNode l1, ListNode l2) {
        // TODO: Write your code here
        return l1;
    }
}
```

Test Case | Run Code Result | Stdout

```
[1,3,5]
[2,4,6]
```

Use Example Testcases | Run | Submit